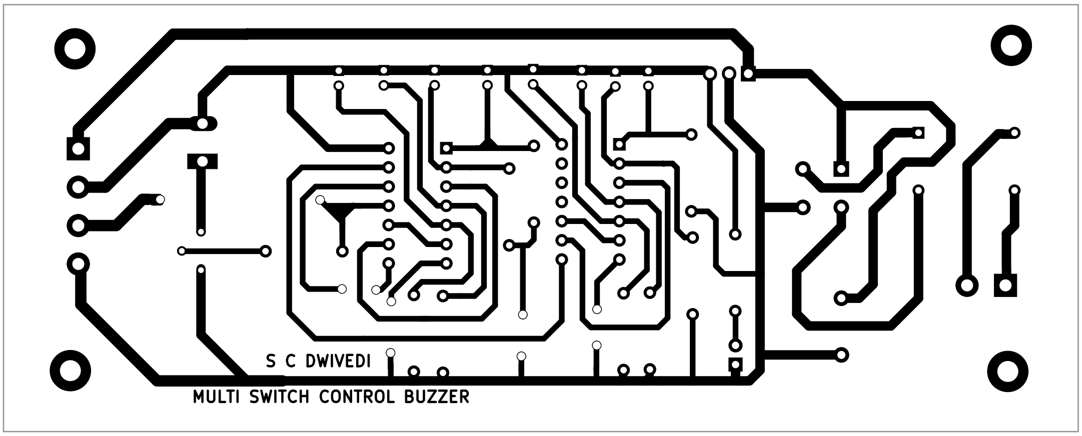

Fig. 3: Actual-size PCB of multi switch control buzzer

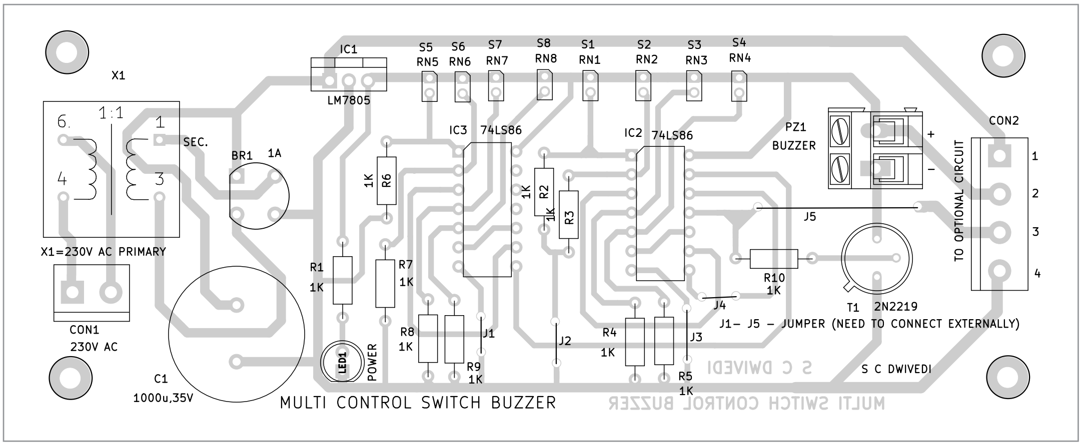

Fig. 4: Component layout of the PCB

TABLE 1
BUZZER ON/OFF / APPLIANCE ON/OFF

S1	S2	S3	S4	S5	S6	S7	S8	Pin11 of IC2D	Buzzer on/off status when switch is pressed	Appliance is on for 1-15 minutes based on VR1 setting
0	0	0	0	0	0	0	0	0	off	off
1	0	0	0	0	0	0	0	1	on	on
0	1	0	0	0	0	0	0	1	on	on
0	0	1	0	0	0	0	0	1	on	on
0	0	0	1	0	0	0	0	1	on	on
0	0	0	0	1	0	0	0	1	on	on
0	0	0	0	0	1	0	0	1	on	on
0	1	0	0	0	0	1	0	1	on	on
0	0	0	0	0	0	0	1	1	on	on

the output of the last XOR gate (IC2D) will be high. Transistor T1 is connected to the output pin 11 of the XOR gate IC2D via resistor R10.

The buzzer is connected between the collector of transistor T1 and 5V DC. When any of the switches (S1 through S8) is pressed, the buzzer gets activated to produce an audible alarm.

The circuit requires 5V DC to operate, which is derived from the 12V, 500mA secondary transformer X1. The 230V AC mains supply is connected to the primary of X1 via CON1 in the circuit. The 12V AC secondary is connected to bridge rectifier BR1 for rectification. The rectified voltage is filtered by capacitor C1 and given to the input of LM7805 to obtain regulated 5V for the circuit's operation. Additionally, it is used to drive relay RL2 used in the optional circuit. The glowing of LED1 indicates the availability of 5V.

The IC 74LS86 contains four individual two-input EX-OR gates. An exclusive OR gate, often abbreviated as EX-OR, performs the logic that the output of a 2-input exclusive OR gate becomes high if one and only one input will be high. For proper understanding of the device, refer to Table 1, which clarifies that the buzzer sounds only when one switch (out of 8) is pressed.

Construction and testing

An actual-size, single-side PCB layout for the multi-switch control buzzer circuit is shown in Fig. 3 and its component layout in Fig.